AWS Big Data Blog

Query Amazon S3 Tables from Amazon EMR Trino using the Iceberg REST endpoint

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Organizations running analytics on Amazon Simple Storage Service (Amazon S3) data lakes often struggle with the operational overhead of managing Apache Iceberg tables, including compaction, snapshot expiration, and metadata tracking, while still needing fast, interactive SQL access across large volumes of data. Amazon S3 Tables, a capability of Amazon S3, addresses this by providing a purpose-built storage layer with native Apache Iceberg support and automated table maintenance. When you query S3 Tables from Amazon EMR using Trino and the Iceberg REST endpoint, you get a fully managed, open-standards-based analytics stack without the undifferentiated heavy lifting of table upkeep.

When paired with Amazon EMR running Trino, organizations gain access to a high-performance distributed SQL query engine capable of processing large-scale datasets. Trino's ability to query data across multiple sources, combined with the automated optimization features of S3 Tables, creates a flexible analytics platform. The integration uses Apache Iceberg's REST catalog specification, providing a standardized interface that supports compatibility across different compute engines while maintaining full control over query execution and data processing logic.

This architectural pattern is particularly valuable for organizations seeking to modernize their data platforms without vendor lock-in, as it relies on open standards and formats. The solution delivers high-throughput query performance with distributed SQL execution while significantly reducing the operational burden of managing table metadata, compaction, and snapshot lifecycle management. In this post, we show you how to create and query Amazon S3 Tables using Trino on Amazon EMR through the Apache Iceberg REST catalog endpoint.

Solution overview

This implementation demonstrates a complete integration between the Trino distribution on Amazon EMR and Amazon S3 Tables through the Apache Iceberg REST catalog endpoint. The architecture uses several key AWS services working in concert:

Amazon EMR serves as the managed compute layer, providing a scalable Hadoop framework that hosts the Trino query engine. Amazon EMR handles cluster provisioning, configuration management, and automatic scaling, allowing teams to focus on analytics rather than infrastructure management.

Apache Trino acts as the distributed SQL query engine, offering ANSI SQL compatibility and the ability to process queries across massive datasets with low latency for interactive workloads. Its connector architecture supports integration with various data sources, including the Iceberg REST catalog.

Amazon S3 Tables provides the storage and catalog layer, managing Apache Iceberg tables with built-in optimization. The service automatically handles compaction, snapshot expiration, and metadata management, reducing operational overhead while maintaining query performance. S3 Tables exposes a REST API endpoint that conforms to the Apache Iceberg REST catalog specification, which provides standardized integration with any Iceberg-compatible engine.

Apache Iceberg REST endpoint serves as the communication protocol between Trino and S3 Tables. This RESTful interface handles catalog operations including namespace management, table creation, metadata retrieval, and transaction coordination. The endpoint supports AWS Signature Version 4 authentication for secure access to table resources.

The data flow follows this pattern: Users submit SQL queries through the Trino CLI or JDBC interface. Trino's Iceberg connector communicates with the S3 Tables REST endpoint to retrieve table metadata and plan query execution. The query engine then reads data directly from S3 using optimized file formats (Parquet, ORC) while using Iceberg's metadata layer for partition pruning and predicate pushdown. Write operations follow a similar path, with Trino coordinating with S3 Tables to commit new data files and update table metadata atomically.

This architecture delivers several key benefits: separation of compute and storage for independent scaling, automated table maintenance reducing operational costs, open-source format compatibility preventing vendor lock-in, and fine-grained access control through AWS Identity and Access Management (IAM) and AWS Lake Formation integration.

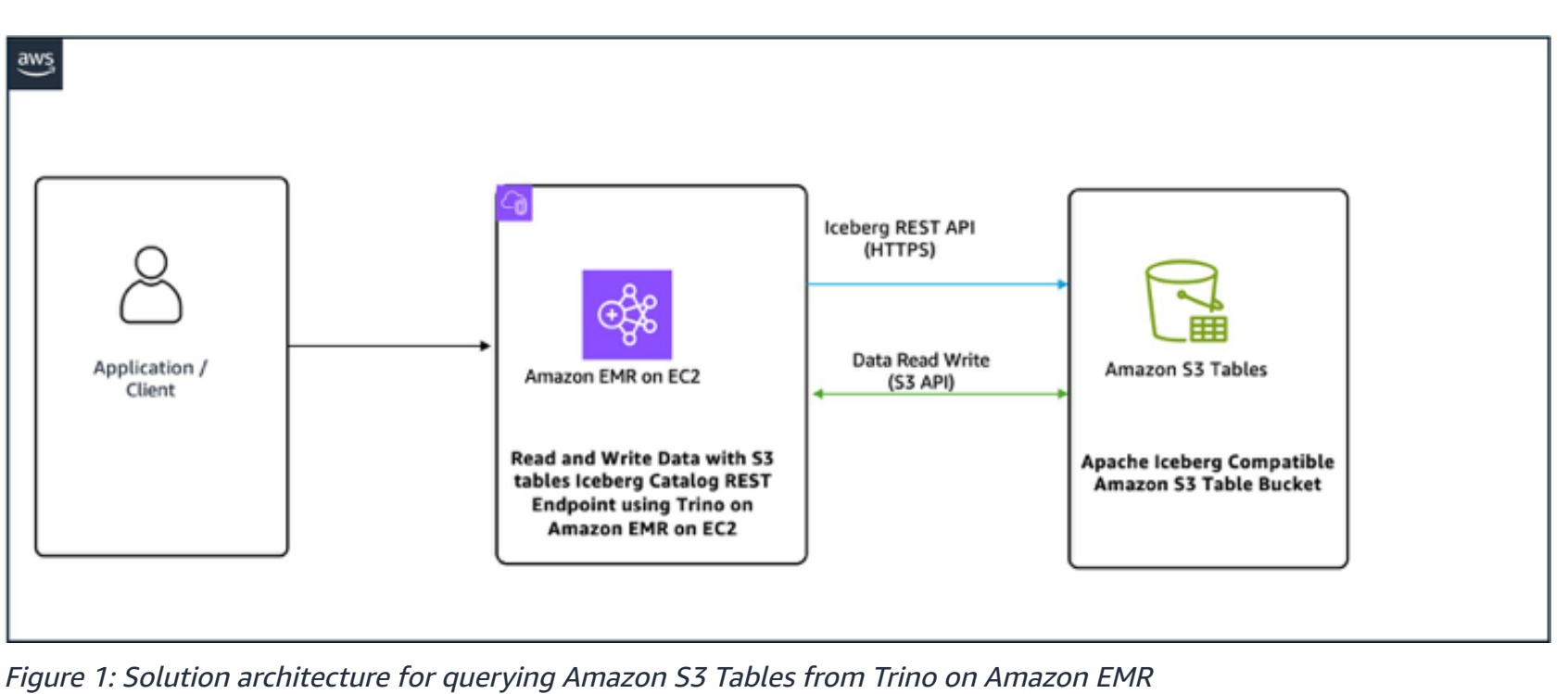

aws
Application / Client
Amazon EMR on EC2
Read and Write Data with S3 tables Iceberg Catalog REST Endpoint using Trino on Amazon EMR on EC2
Iceberg REST API (HTTPS)
Data Read Write (S3 API)
Amazon S3 Tables
Apache Iceberg Compatible Amazon S3 Table Bucket

Figure 1: Solution architecture for querying Amazon S3 Tables from Trino on Amazon EMR

Prerequisites

Before getting started, make sure that you have the following:

- An active AWS account with billing enabled.
- An AWS Identity and Access Management (IAM) user with specific permissions to create and manage resources, such as a virtual private cloud (VPC), subnet, security group, IAM roles, Amazon EMR, Interface VPC endpoints, S3 Tables bucket and S3 buckets.
- Sufficient VPC capacity in your chosen AWS Region.

For this post, we create the solution resources in the US East (N. Virginia) Region (`us-east-1`) using AWS CloudFormation templates. In the following sections, we show you how to configure your resources and implement the solution.

Note: Querying Amazon S3 Tables through Trino on Amazon EMR requires Trino version 475 or later, available in Amazon EMR 7.11 and later.

Part A: Configure Amazon S3 Tables integration with Trino on Amazon EMR using AWS CloudFormation

In this post, you use the CloudFormation template `emr-trino-s3tables.yaml` .

- This template deploys the following resources: a VPC with one private subnet, an S3 Tables interface VPC endpoint for private access, and an Amazon EMR cluster running Trino integrated with Amazon S3 Tables through the Apache Iceberg REST catalog endpoint.
- It also creates an S3 Tables bucket, a general-purpose S3 bucket, IAM roles, and security groups.
- At deploy time, it dynamically generates the Trino catalog configuration and bootstrap script.

To create the solution resources, complete the following steps:

1. Launch the stack `emr-trino-s3tables.yaml` using the CloudFormation template.

1. Provide the parameter values as listed in the following table.

Parameters	Description	Sample value
Stack Name	Name of CloudFormation stack	emr-s3tables-trino
VPC CIDR block	IP range (CIDR notation) for this VPC.	10.0.0.0/16
Private Subnet CIDR block	IP range (CIDR notation) for the private subnet in the second Availability Zone.	10.0.1.0/24
Resource name Prefix	Short prefix applied to every resource name	emr-s3tables
S3 Tables bucket name	Name of S3 table Bucket	trinoemrs3tablebuck
EMR release	Release version of Amazon EMR	EMR 7.12

The stack creation process can take approximately 15 minutes to complete. You can check the Outputs tab for the stack after the stack is created, as shown in the following screenshot.

Figure 3: CloudFormation stack outputs

Figure 3: CloudFormation stack outputs

Understanding the deployment

The CloudFormation template performs several key tasks:

1. **Infrastructure provisioning**: Sets up the Amazon EMR cluster with Trino, VPC, subnet, security group, and S3 table bucket.
2. **Configuration**: Creates necessary Trino configuration files.
3. **Integration** configuration: Sets up the Iceberg REST connector for S3 Tables.

Part B: Connecting Trino to Amazon S3 Tables with Iceberg REST endpoint

The CloudFormation template automatically configures the S3 Tables catalog in Trino on Amazon EMR. In the next section, we examine the configuration that drives this integration.

1. Catalog configuration details

A catalog in Trino on Amazon EMR is the configuration that grants access to a specific data source. Each Trino on Amazon EMR cluster can have multiple catalogs configured, allowing access to different data sources simultaneously.

As part of this setup, the CloudFormation template creates a catalog properties file at `/etc/trino/conf/catalog/s3tables_irc.properties` with the following configuration:

```
connector.name=iceberg
iceberg.catalog.type=rest
iceberg.rest-catalog.uri=https://s3tables.<REGION>.amazonaws.com/iceberg
iceberg.rest-catalog.warehouse=arn:aws:s3tables:AwsRegion:<ACCOUNT-ID>:bucket/<BUCKET-NAME
iceberg.rest-catalog.sigv4-enabled=true
iceberg.rest-catalog.signing-name=s3tables
iceberg.rest-catalog.view-endpoints-enabled=false
fs.hadoop.enabled=false
fs.native-s3.enabled=true
s3.region=us-east-1
s3.iam-role=arn:aws:iam::<ACCOUNT-ID>:role/service-role/<ROLE-NAME>
```

2. S3 Tables Iceberg REST endpoint configuration properties

The following table lists the key properties in the catalog configuration on Trino:

Property name	Description
iceberg.rest-catalog.uri	REST server API endpoint URI (necessary).
iceberg.rest-catalog.warehouse	Warehouse ID or location for the catalog (necessary). For S3 Tables, this is the ARN for the S3 table bucket as shown in the preceding properties example.
iceberg.rest-catalog.sigv4-enabled	Must be set to 'true' (necessary)
iceberg.rest-catalog.signing-name	Must be set to 's3tables' (necessary)
iceberg.rest-catalog.view-endpoints-enabled	Must be set to 'false' (necessary)
fs.hadoop.enabled	Must be set to 'false'
fs.native-s3.enabled	Must be set to 'true'
s3.iam-role	Amazon Resource Name (ARN) of the IAM role with permissions to S3 Tables. In this post, we use the same role, which is the service role for Amazon EMR.
s3.region	AWS Region, for example us-east-1

This configuration establishes a connection between Trino and the S3 Tables REST endpoint. You can have multiple catalogs registered, one per S3 table bucket, which is determined by the `iceberg.rest-catalog.warehouse` property.

3. Configure Amazon EMR service IAM role trust relationships

The Amazon EMR service role requires proper trust relationships to function correctly. Navigate to the IAM console and configure the trust policy for your Amazon EMR service role:

```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Effect": "Allow",
            "Principal": {
                "Service": "elasticmapreduce.amazonaws.com"
            },
            "Action": "sts:AssumeRole"
        },
        {
            "Effect": "Allow",
            "Principal": {
                "AWS": "arn:aws:iam::<ACCOUNT-ID>:role/service-role/AmazonEMR-InstancePr
            },
            "Action": "sts:AssumeRole"
        }
    ]
```

This trust policy establishes two critical relationships:

1. The Amazon EMR service can assume the role to manage cluster operations.
2. The EC2 instance profile can assume the role to access S3 Tables with elevated permissions.

4. Working with S3 Tables in Trino on Amazon EMR

Now that you have Trino on Amazon EMR set up and configured to work with S3 Tables, you can explore how to work with this integration.

4.1. Connecting to Trino on Amazon EMR

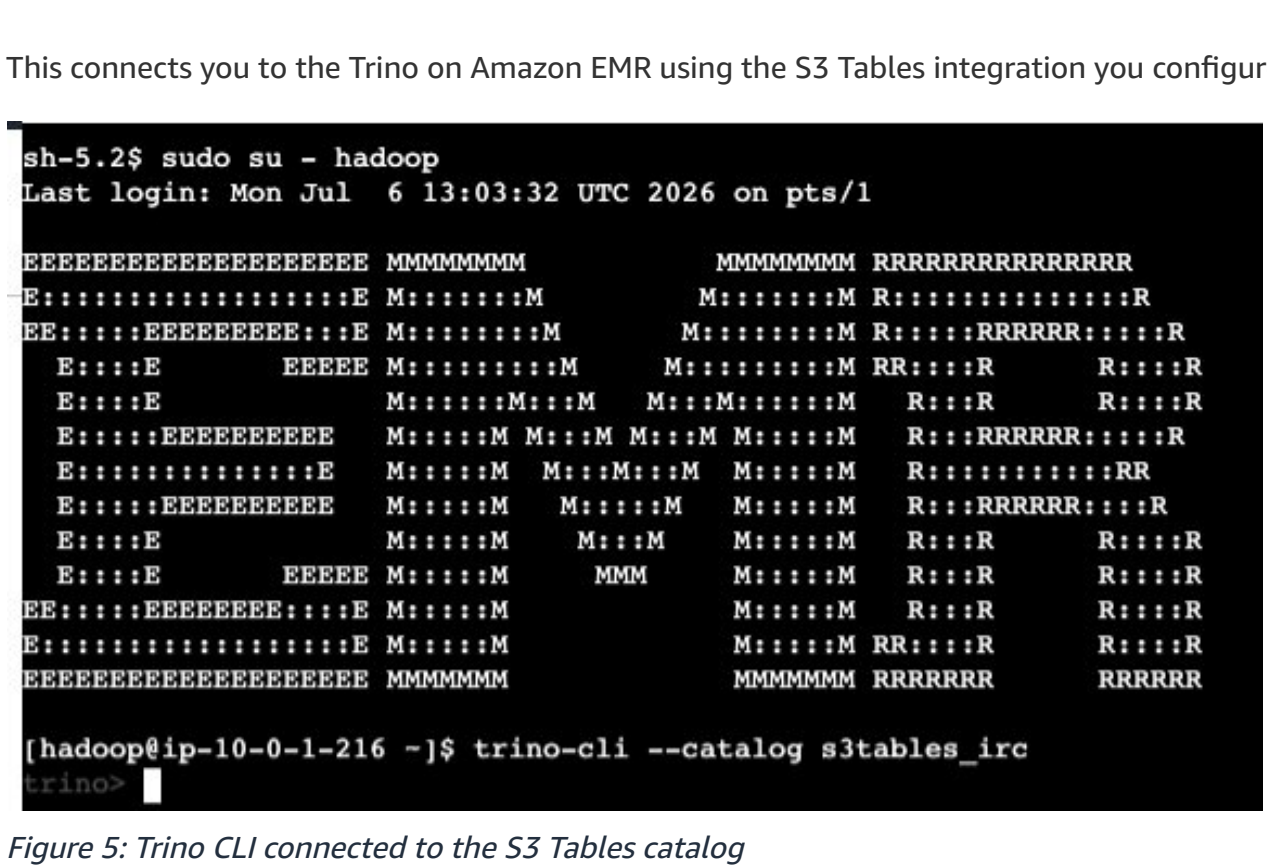
Navigate to Amazon EMR and select Connect to the primary node using AWS Systems Manager Session Manager for passwordless SSH.

Figure 4: Connecting to the primary node with Session Manager

When you're connected, you can use the Trino CLI with your S3 Tables catalog:

```
sudo su - hadoop
trino-cli --catalog s3tables_irc
```

This connects you to the Trino on Amazon EMR using the S3 Tables integration you configured.



```
sh-5.2$ sudo su - hadoop
Last login: Mon Jul  6 13:03:32 UTC 2026 on pts/1

EEEEEEEEEEEEEEEEEEEE MMMMMMMM           MMMMMMMM RRRRRRRRRRRRRRR
E::::::::::::::::::E M:::::::M         M:::::::M R::::::::::::::R
EE:::::EEEEEEEEE:::E M::::::::M       M::::::::M R:::::RRRRRR:::::R
  E::::E       EEEEE M:::::::::M     M:::::::::M RR::::R      R::::R
  E::::E             M::::::M:::M   M:::M::::::M   R:::R      R::::R
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EEEEEEEEEEEEEEEEEEEE MMMMMMM             MMMMMMM RRRRRRR      RRRRRR

[hadoop@ip-10-0-1-216 ~]$ trino-cli --catalog s3tables_irc
trino>
```

Figure 5: Trino CLI connected to the S3 Tables catalog

4.2. Examples: Creating and querying tables

In this section you run through some example queries to demonstrate the functionality.

4.2.1 Creating a namespace

First, you create a namespace (schema) in S3 Tables. A namespace in S3 Tables is a logical container or organizational unit that helps group related tables and objects together.

```
CREATE SCHEMA blog_namespace;
USE blog_namespace;
```

4.2.2 Creating a table

Create a table with various data types. You don't need to specify the table type as Iceberg explicitly because you're connecting to the Iceberg catalog. You can use all standard Iceberg capabilities, such as partitioning and sorting. Furthermore, some of the important Iceberg table properties that support table maintenance operations are configured with default values. You also have the option to edit the configurations using S3 Tables maintenance APIs.

```
CREATE TABLE IF NOT EXISTS customers (
customer_sk INT,
customer_id VARCHAR,
salutation VARCHAR,
first_name VARCHAR,
last_name VARCHAR,
preferred_cust_flag VARCHAR,
birth_day INT,
birth_month INT,
birth_year INT,
birth_country VARCHAR,
login VARCHAR
) WITH (
format = 'PARQUET',
sorted_by = ARRAY['customer_id']
);
```

Table property explanation:

- `format = 'PARQUET'` : Specifies Parquet as the file format for optimal compression and query performance.
- `sorted_by = ARRAY['customer_id']` : Defines sort order within data files, improving query performance for `customer_id` filters.

Verify the table creation:

```
SHOW TABLES;
```

You should see `customers` in the output, confirming the table exists in the S3 Tables catalog.

4.2.3 Inserting data

You can insert some sample data into your table. You can also use an existing table in any of the catalogs configured in Trino on Amazon EMR to read data and write into the S3 table with an `INSERT INTO ... SELECT` statement.

```
INSERT INTO customers VALUES
(1, 'AAAAA', 'Mrs', 'Martha', 'Rivera', 'Y', 8, 4, 1984, 'US', 'mrivera'),
(2, 'AAAAB', 'Mr', 'Mateo', 'Jackson', 'N', 22, 6, 2001, 'US', 'mjackson'),
(3, 'BAAAA', 'Ms', 'Mary', 'Major', 'Y', 16, 2, 1999, 'US', 'mmajor'),
(4, 'BBAAA', 'Mr', 'Paulo', 'Santos', 'N', 30, 3, 1973, 'US', 'psantos'),
(5, 'AACAA', 'Ms', 'Ana', 'Silva', 'N', 2, 6, 1982, 'CA', 'asilva'),
(6, 'ABAAA', 'Mr', 'Alejandro', 'Rosalez', 'N', 5, 12, 1988, 'US', 'arosalez'),
(7, 'BBAAA', 'Ms', 'Nikki', 'Wolf', 'N', 6, 1, 2006, 'MX', 'nwolf'),
(8, 'ACAAA', 'Mr', 'Arnav', 'Desai', 'N', 15, 7, 1976, 'US', 'adesai');
```

This INSERT operation demonstrates Trino's ability to write data to S3 Tables. Behind the scenes, Trino:

1. Writes data files in Parquet format to S3.
2. Communicates with the S3 Tables REST endpoint to register the new files.
3. Atomically commits the transaction, updating table metadata.

4.2.4 Querying data

Execute a `SELECT` query to retrieve and verify the inserted data:

```
SELECT * FROM customers LIMIT 10;
```

The query should return all eight customer records with proper formatting. You can also execute more complex analytical queries:

```
-- Count customers by country
SELECT birth_country, COUNT(*) as customer_count
FROM customers
GROUP BY birth_country
ORDER BY customer_count DESC;

-- Find customers born after 1990
SELECT first_name, last_name, birth_year
FROM customers
WHERE birth_year > 1990
ORDER BY birth_year;
```

These queries demonstrate Trino's SQL capabilities and the integration with S3 Tables for both read and write operations.

4.3 Explore advanced features

S3 Tables with Iceberg provides several features for data management:

4.3.1 Time travel queries

Step 1: Check available snapshots.

```
-- Query table as of a specific timestamp. Check available snapshots
SELECT * FROM "customers$snapshots";
```

Step 2: Query the table as of a specific snapshot.

```
SELECT * FROM customers FOR VERSION AS OF <snapshot_id_from_step1>;
```

4.3.2 Schema evolution

```
-- Add a new column
ALTER TABLE customers ADD COLUMN email VARCHAR;

-- Rename a column
ALTER TABLE customers RENAME COLUMN login TO username;
```

Cleaning up

To clean up the resources, navigate to CloudFormation and delete the stack that you created.

Conclusion

This solution demonstrates an integration between Amazon EMR Trino and Amazon S3 Tables using the Apache Iceberg REST catalog specification. In this post, we showed you how to create and query S3 Tables from Trino on Amazon EMR. The architecture delivers several advantages for modern data platforms:

Operational simplicity: S3 Tables eliminates the complexity of managing Iceberg table metadata, compaction schedules, and snapshot lifecycle policies. The service handles these operations automatically, allowing data teams to focus on analytics rather than infrastructure maintenance.

Performance at scale: The architecture is designed for large-scale workloads. Trino distributes query execution across the cluster while Iceberg's metadata layer helps the engine locate only the relevant data files. Features like partition pruning, predicate pushdown, and columnar file formats can help improve performance for both interactive and batch workloads.

Cost efficiency: This architecture separates compute and storage, so you can scale each independently based on workload requirements. S3 Tables automatically compacts small files to help reduce storage overhead, and Amazon EMR clusters can scale dynamically so you pay for compute only when needed.

Open standards and portability: By using Apache Iceberg's open table format and REST catalog specification, this solution avoids vendor lock-in. Other Iceberg-compatible engines can access tables created in S3 Tables including Apache Spark, Apache Flink, and Dremio, providing flexibility in tool selection.

Fine-grained access control: Integration with IAM and resource-based policies provides access control at the table bucket, namespace, and table level. For fine-grained access at the column and row level, you can integrate with AWS Lake Formation. AWS Signature Version 4 authentication supports secure communication between Trino and S3 Tables.

ACID transactions: Iceberg's transaction model guarantees atomicity, consistency, isolation, and durability for all table operations. This supports reliable concurrent reads and writes, making the platform suitable for production workloads requiring data consistency.

This architectural pattern is particularly well-suited for organizations building modern data lakehouses, migrating from traditional data warehouses, or consolidating multiple analytics platforms. The combination of the managed compute of Amazon EMR, Trino's versatile query engine, and the automated table management of S3 Tables creates a strong foundation for data-driven decision making.

To learn more about the services and features discussed in this post, see the following resources:

- Amazon EMR
- Getting started with Amazon S3 Tables
- Trino on Amazon EMR

About the authors

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